

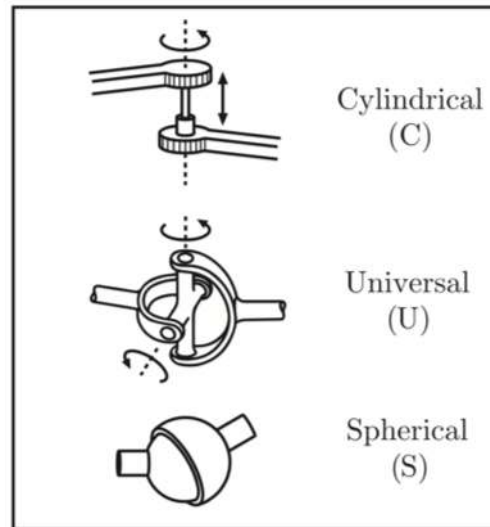
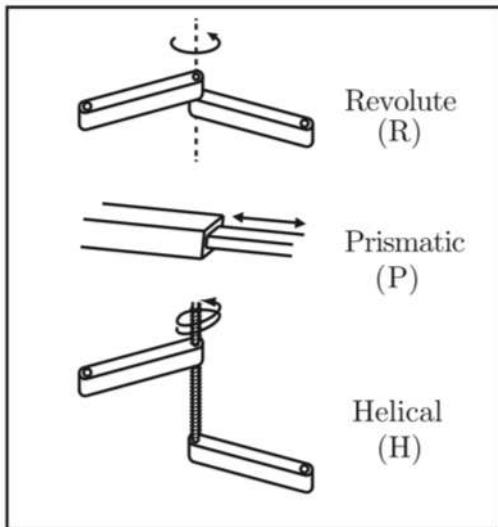
Chapter 2	Configuration Space
	2.1 DOF of a Rigid Body
	2.2 DOF of a Robot
Chapter 3	Rigid-Body Motions
Chapter 4	Forward Kinematics
Chapter 5	Velocity Kinematics and Statics
Chapter 6	Inverse Kinematics
Chapter 7	Kinematics of Closed Chains
Chapter 8	Dynamics of Open Chains
Chapter 9	Trajectory Generation
Chapter 10	Motion Planning
Chapter 11	Robot Control
Chapter 12	Grasping and Manipulation
Chapter 13	Wheeled Mobile Robots

Important concepts, symbols, and equations

- **configuration**: a specification of the positions of all points of a mechanism
- **degrees of freedom** (dof): # of real #s required to describe a configuration
- **configuration space** (C-space): the dof-dimension space of all configurations
- dof of a planar body: $m = 3$; dof of a spatial body: $m = 6$

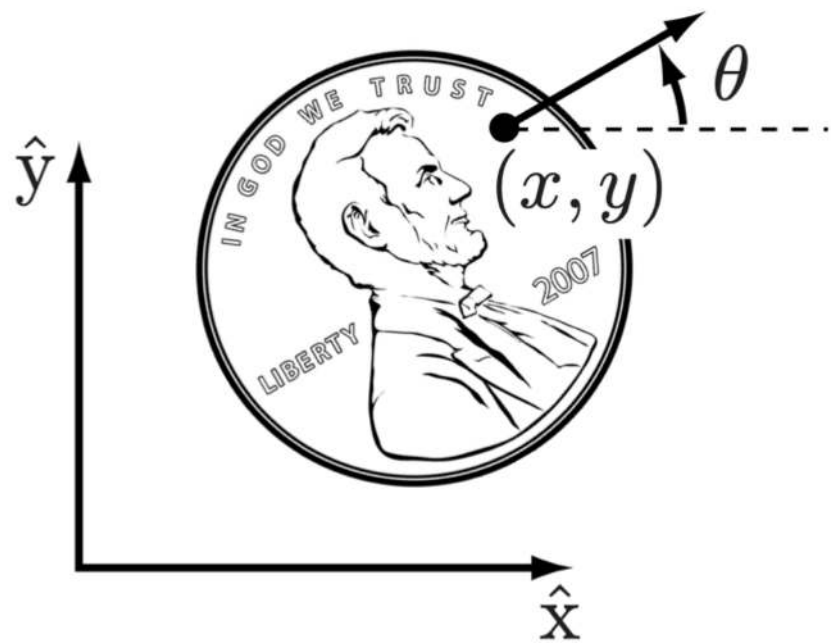
Important concepts, symbols, and equations (cont.)

- mechanism dof = Σ (body freedoms) – Σ (independent constraints from joints)
- joint types:

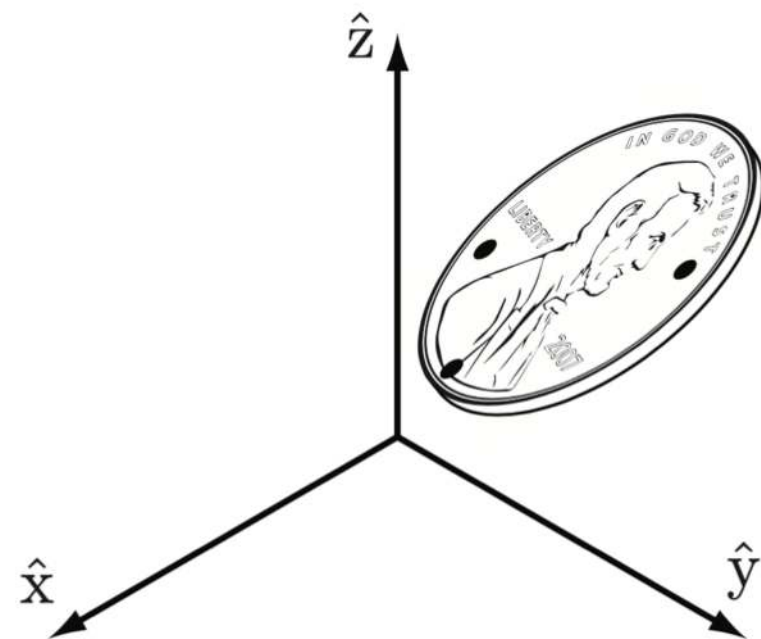


Joint type	dof f	Constraints c between two planar rigid bodies	Constraints c between two spatial rigid bodies
Revolute (R)	1	2	5
Prismatic (P)	1	2	5
Helical (H)	1	N/A	5
Cylindrical (C)	2	N/A	4
Universal (U)	2	N/A	4
Spherical (S)	3	N/A	3

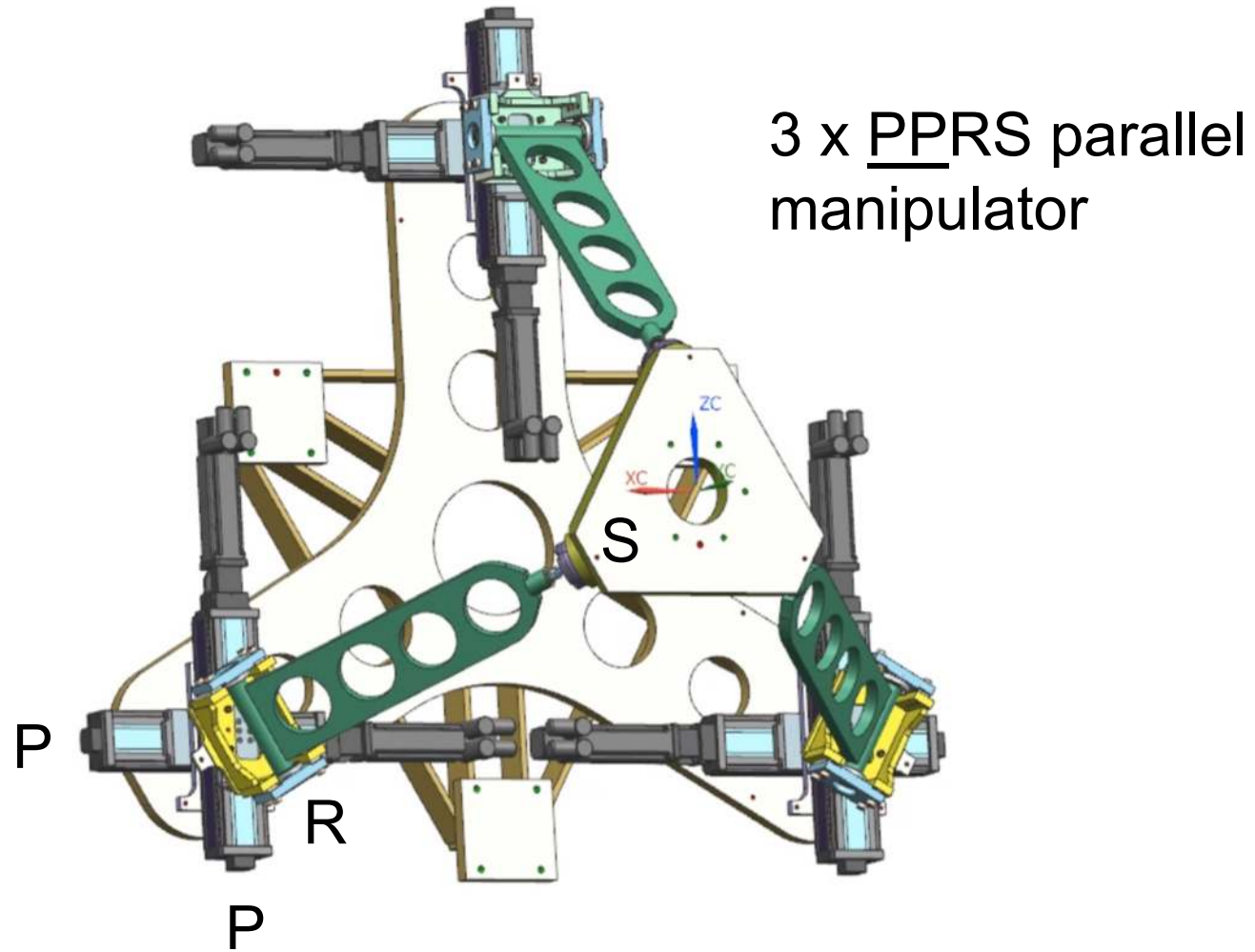
- Grübler's formula:**
$$\text{dof} = m(N - 1 - J) + \sum_{i=1}^J f_i$$



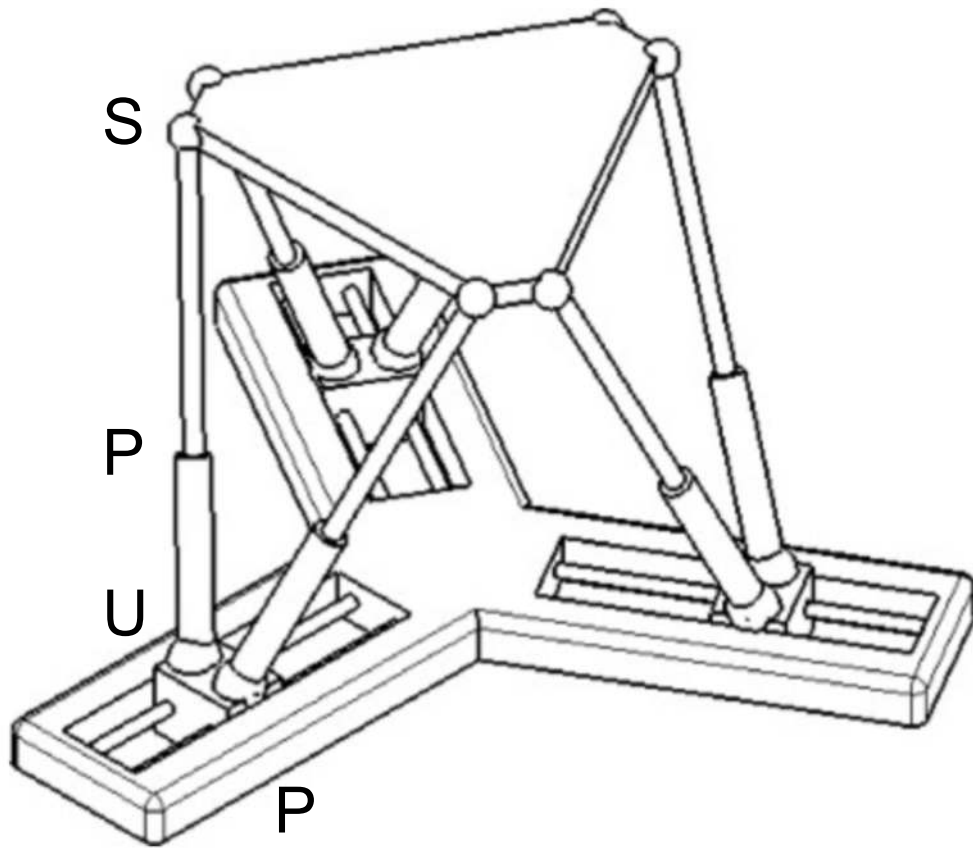
coin on a plane



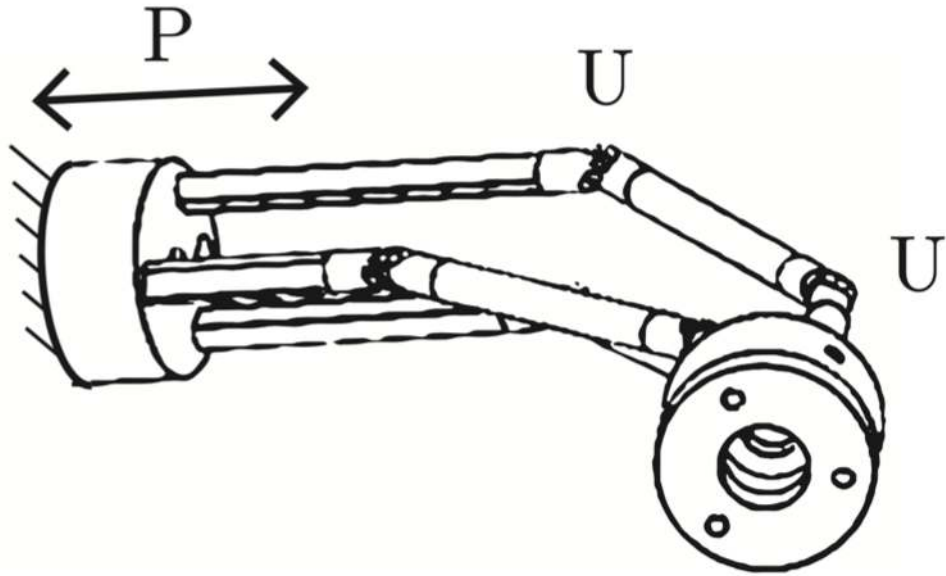
coin in space



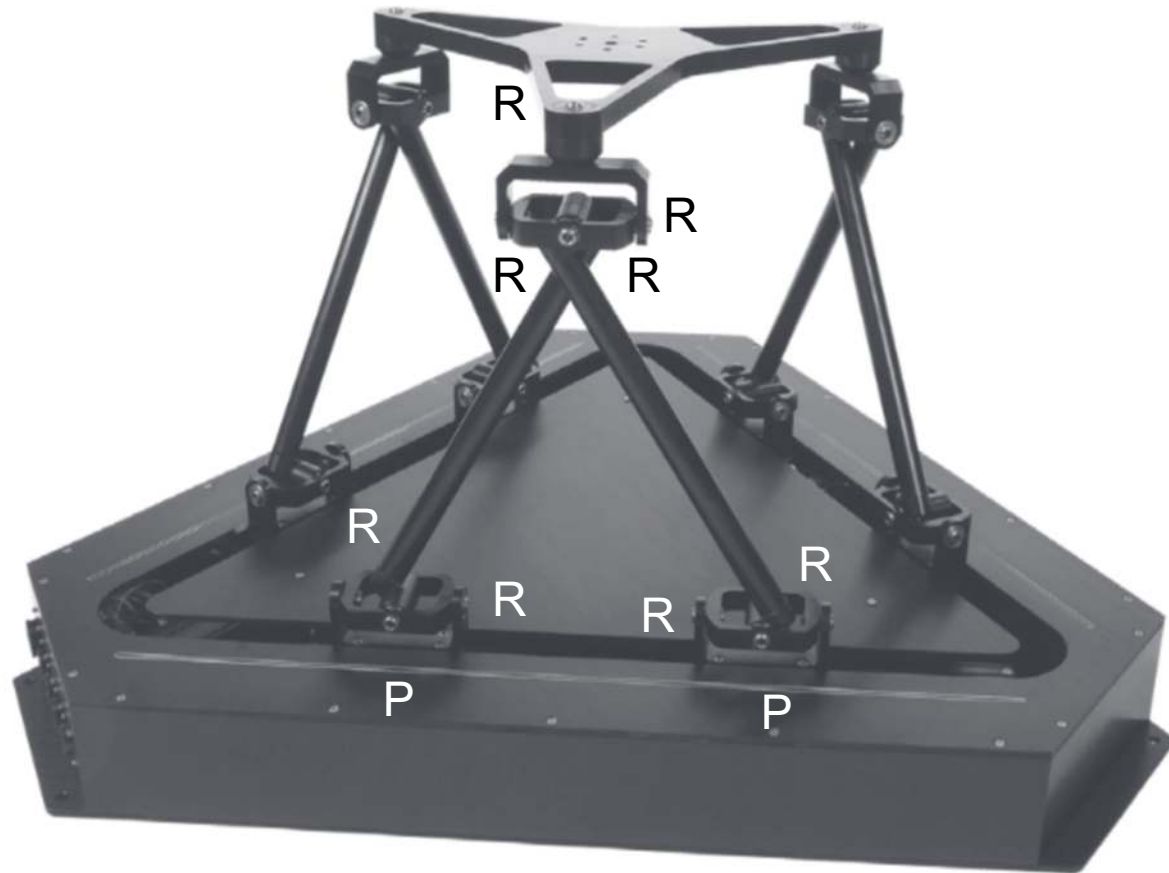
KUKA Systems North America LLC
(patent pending)



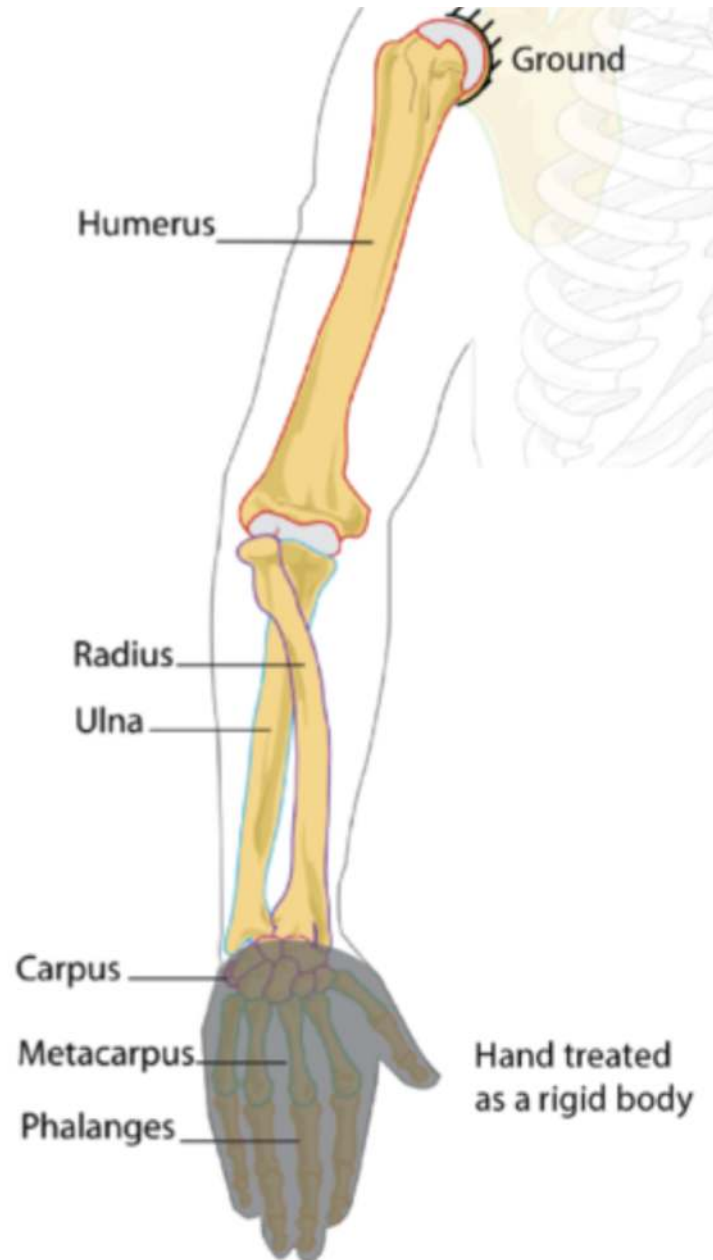
3 x $\underline{P}$ UU miniature surgical parallel manipulator (National University of Singapore)



Quanser Hexapod



<https://www.youtube.com/watch?v=AyVu4AE25DM>



How many dof does the human arm have?

Method 1: add dof of joints (shoulder, elbow, wrist)

Method 2: fully constrain hand's position

How many total constraints are imposed by the joints?